

RNA Digital: Improving marketing campaign ROI with real-time monitoring

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ABOUT RNA Digital

With Google Cloud, RNA Digital is delivering a scalable, near-100% available marketing monitoring platform that enables enterprises to monitor their customers to minimize disruption from technical issues or anomalies.



About RNA Digital

Headquartered in Sydney and a team of 10, RNA Digital enables businesses to optimize marketing spend by identifying issues such as API errors in near real-time. As a result, those businesses can take immediate corrective action to minimize the impact on revenue and customer experience.

Industries: Technology

Google Cloud results

- Supports millions of requests per hour from enterprise customers.
- Alerts customers 10,080x faster so customers can respond almost immediately
- Enables the business to add significant new functionalities in just three weeks

Delivered
10,080x
improvement
the time to
detect

Location: Australia

(<https://cloud.google.com/kubernetes-engine/>)

Google Cloud (<https://cloud.google.com/>)

App Engine (<https://cloud.google.com/appengine/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Google Cloud Armor
(<https://cloud.google.com/armor/>)

Dataflow (<https://cloud.google.com/dataflow/>)

Kubernetes Engine
(<https://cloud.google.com/kubernetes-engine/>)

Growing sales, increasing brand awareness and generating leads are essential to business success, so digital marketing campaigns must be targeted, well-executed and tracked in detail. However, the pieces of code, also known as tags, that businesses place on their brand websites to track user behavior and other data often fail and experience anomalies.

If a business does not immediately see and respond to these events, its ad operations and analytics may be compromised, and its ad spend wasted.

For Ragy Abraham and Andrew Nada, solving this problem presented an exciting opportunity. With combined experience spanning web analytics, data science, consulting, they were well equipped to address the opportunity.

With developer Peter Ebeid, the pair founded RNA Digital (<https://www.rna.digital/>). The Sydney-headquartered business's flagship product is Monita, an observability platform that runs on Google Cloud. Monita provides real-time monitoring of every tag fired when its trigger is met, by device, browser, and operating system, so when the platform detects a failure or anomaly, it can alert a customer immediately by email or other communications service.

"Whenever businesses deploy ads these days, they need accurate signals about how much their customers spend and more to determine whether their campaigns are successful or not," says Abraham. "Monita enables brands and agencies to spot signal downtime and remediate them in near real-time."

Monita now incorporates deep synergies with the Google Ads ecosystem, with Abraham pointing to the platform's integration with Google Tag Manager as pivotal to its value proposition.

As a SaaS startup, RNA Digital established its priorities as speed to market and scaling cost-effectively, without imposing an excessive administration burden on its small team. "As a startup, we needed a scalable, functional, and reliable product in-market as quickly as possible to establish ourselves," explains Abraham. "Without the resources to manage infrastructure, we saw fully managed services like Google Cloud (<https://cloud.google.com/>) as the answer to our constraints."

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Working with a provider that can deliver infrastructure economies

RNA Digital reviewed public cloud services and identified Google Cloud as providing the required infrastructure economies and services the business could integrate into an efficient design. Through the Google Cloud startup program, RNA Digital also had access to credits that could turbocharge the business. "Google Cloud also had an intuitive interface, user-friendly service naming conventions and consistent, thorough documentation," says Abraham. "Beyond these attributes, we admired how Google Cloud constantly innovated in developing new cloud products, and its rich relationship with the open source community."

RNA Digital opted to go 'all-in' on Google Cloud in 2019 and started with services that, according to Abraham, provided "the least friction from ideation to deployment" to get up and running quickly, including building applications with [App Engine](#)

(<https://cloud.google.com/appengine>). As the business's experience and confidence with the cloud platform improved, its appetite for more cloud services increased. RNA Digital has now matured to take full ownership of its cloud environment, with Google Kubernetes Engine (<https://cloud.google.com/gke>) (GKE) operating as its core infrastructure, BigQuery (<https://cloud.google.com/bigquery>) as its data warehouse, and Dataflow (<https://cloud.google.com/dataflow>) for data pipelines.

"I don't think we could replicate the real-time processing we do in Dataflow elsewhere," explains Abraham. "Because Dataflow has unparalleled scale and integrates across so many platforms, we can undertake complex data manipulation and enrichment. Also, we can write pipelines in a range of languages, with the option of multi-language pipelines." The business routes the data originating from customer websites through these pipelines into BigQuery for storage, query and analysis.

With Dataflow, RNA Digital has been able to generate data validation alerts in just one second, a 10,080x improvement on its previous capability. This improvement enables businesses to, for example, react quickly to a notification of an API error that compromises customers' ability to purchase their products and services, and fix the issue before it impacts revenue and reputation.

Meanwhile, running on GKE gives RNA Digital the infrastructure flexibility the business needs to adapt seamlessly to changing customer usage. "Google Kubernetes Engine provides best-in-class

infrastructure orchestration and enables us to be very cost-effective, because it scales up and down based on demand and we are only charged for what we use," says Abraham.



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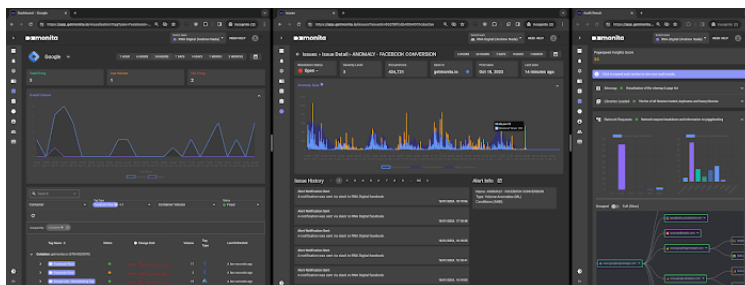
Delivering an SLA with Google Cloud that wins new customers

With GKE, Monita has not experienced any infrastructure related downtime to date, helping RNA Digital market and deliver an SLA of 99.99 percent that meets or exceeds customers' strict requirements. This figure is boosted by the team's ability to push out an update without incurring downtime. "When signing up a new enterprise customer recently, we stressed this SLA extensively in our master services agreement," says Abraham, adding that having such confidence in uptime made all the difference in customer retention.

RNA Digital is now using Monita to provide services to a range of high profile enterprise customers that, collectively, send millions of requests per hour to the platform, comprising data around user behavior, site conversions, web traffic and more.

Beyond these core services, RNA Digital uses a range of Google Cloud offerings to address issues that have grown in importance as the business matures, such as security. For example, Google Cloud Armor (<https://cloud.google.com/armor>) enables the business to protect itself from distributed denial-of-service (DDoS) and application attacks that could otherwise compromise its offering and reputation among new and existing clients.

With Google Cloud, flexibility and agility is now the mantra at RNA Digital, with the business rolling out new features and capabilities quickly to enhance the customer experience. For example, when a customer asked for a compliance dashboard to show all instances when tags/pixels send personally identifiable information (PII) breaches in Monita, RNA Digital was able to fulfill the request in just two weeks. The result was that customers were able to meet their compliance obligations, eliminating all instances of PII breaches generated from their websites.



Maintaining an application with low overheads and minimal pressure

Working with Google Cloud has also enabled RNA Digital's engineering team to rest assured knowing there are multiple redundancies built into the platform.

Abraham describes the fact RNA Digital has been able to maintain Monita with low operational overheads and minimal pressure on its internal team as "a measure of success."

"Thanks to Google Cloud, we've been able to maintain Monita with remarkably fewer resources than typically required for a platform of this scale. This is largely due to our integrated deployment cycle and use of managed services with 99.99 percent uptime," says Abraham.

RNA Digital has made the [Monita platform](https://www.getmonita.io/) (https://www.getmonita.io/) available on Google Marketplace to increase its exposure to prospective customers. "Google Cloud reduces the time required to onboard new enterprise customers as it decreases procurement overhead," says Abraham. "We expect to reduce the procurement cycle from three months to less than one month."

With RNA Digital poised to move to a new level of success, Abraham remains extremely grateful for the support from Google Cloud. "Without Google Cloud, it would have been significantly more difficult to build our business," he concludes. "Other than the platform itself and the services we use, the support from Google Cloud through credits has been a huge help particularly in the early days, and we look

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